

Unit 10: Case Study

DOS

DOS is an abbreviation of Disk Operating System. It was popular operating system software during 1980-1990. DOS is the first operating system used by IBM-compatible computers. It was available in two versions that were essentially the same, but marketed under two different name. PC-DOS is the version developed by IBM and sold to first IBM compatible manufacturers. MS-DOS is the version that Microsoft bought the rights to, and was bundled with the first versions of Windows.

Features of DOS

- 1) It is primitive type of OS.
- 2) It is a 16-bit OS, developed for personal computers.
- 3) It uses CUI, so user needs to provide instructions through the text based commands.
- 4) It uses two types of command: external commands and internal commands.
- 5) It doesn't provide graphics facility but some application programs which run under DOS may provide graphics facility.
- 6) It is a single user/program OS and doesn't supports multi-user and multitasking. In a uni programming system like DOS, time spent waiting for I/O is wasted and CPU is free during this time.
- 7) Monolithic kernel is used.
- 8) DOS uses single contiguous allocation memory management technique.
- 9) DOS uses FAT (File Allocation Table) file system.
- 10) The file name under DOS must not be more than 11 characters long, with 8 character primary name and a 3 character file extension, separated by dot (.).
- 11) DOS provides minimum (lowest level) of security.
- 12) Used on the same computer, DOS is much faster than Windows OS.
- 13) DOS is available in different variations, like MS-DOS, PC-DOS on the basis of different developers.
- 14) MS DOS comprises three essential files for booting the computer. These are:
 - IO.SYS (handle input/output process)
 - MSDOS.SYS (manage workspace for applications)
 - COMMAND.COM (command interpreter- receive and execute commands)

Windows

MS Windows is a GUI (Graphical User Interface) based operating system produced by Microsoft Corporation. Nowadays, Windows OS is used in personal computers, including home and business desktops, laptops, notebooks, media centers (specially designed to focus on multimedia aspects). Almost 90% of all personal computers use this system. MS Windows provides GUI, desktop system, virtual memory management, control panel, multitasking, and support for many peripheral devices.

The versions of Windows developed are Microsoft Interface Manager and Windows 1.0, Windows 2.0, Windows/386, Windows 3.0, Windows 3.1, Multimedia Windows, Windows 3.11, Windows for Workgroups 3.1 & 3.11, Windows for Pen Computing 3.1, Windows 95, Windows NT (New Technology), Windows 98, Windows 2000, Windows Advanced Server 2000, Windows ME, Windows XP, Windows Server 2003, Windows Vista, Windows Server 2008, Windows 7, Windows 8, Windows 10.

Features of Windows

- 1) It is a GUI based OS.
- 2) It uses graphics, colors, icons, folders, desktop, and multimedia making it more users friendly.
- 3) Advanced IO devices such as a mouse, joystick, trackball, track pad, a touch screen can be used.
- 4) Present Windows OS supports multitasking, multi-user, and multiprogramming environment.
- 5) It is 32 or 64 bit OS.
- 6) It uses hybrid kernel.
- 7) Windows used non-preemptive scheduling up to Windows 3.x, and started using pre-emptive scheduling with Win95. Windows uses a round-robin technique with a multi-level feedback queue for priority scheduling ever since NT.
- 8) Two primary Windows mechanisms for IPC are the anonymous pipe and the named pipe, both of which are accessed with the familiar ReadFile and WriteFile functions.
- 9) Windows uses demand paging with clustering. Clustering handles page faults by bringing in not only the faulting page but also the multiple pages surrounding the faulting page. It uses clock algorithm.
- 10) File systems used in Windows are: FAT, FAT32, NTFS (New Technology File System), and exFAT.
- 11) The Windows kernel-mode I/O manager manages the communication between applications and the interfaces provided by device drivers.
- 12) Windows security policies are very effective in protecting the windows machines by providing restricted access to the users. It includes: security settings for Active Desktop, Computer, Control Panel, Explorer, Internet Explorer, Network, and System categories can be defined using Security Policies Configuration.

UNIX

UNIX is a powerful, flexible, and manageable OS with good utilities and communication abilities. It was used in bigger size machines like the mainframe and minicomputers. It was developed by AT&T Bell Labs around 1970. The development of UNIX has given birth to many operating systems that can work on different types a size of computers. For example, Linux and Solaris OS are developed from UNIX. UNIX is a flexible operating system and can be used for any type of processor. It can work on single and multiple processors. UNIX is developed in 'C' and assembly language. Many of the popular commercial operating systems e.g. Linux, Xenix, HP UNIX, SUN OS. has inherited its main features.

Features of UNIX:

- 1) It is a multiuser, multitasking, and multiprocessing operating system.
- 2) It is CUI based.
- 3) Monolithic kernel is used.
- 4) UNIX uses two categories of processes: system processes and user processes. System processes run in kernel mode and execute operating system code to perform administrative and housekeeping functions, such as allocation of memory and process swapping. User processes operate in user mode to execute user programs and utilities and in kernel mode to execute instructions that belong to the kernel.
- 5) The traditional method of inter-process communication in UNIX is the pipe.
- 6) Round robin with multilevel feedback scheduling algorithm is used.
- 7) UNIX memory management scheme includes swapping and demand paging.
- 8) A global page replacement policy and the clock (second chance) algorithm is used.

- 9) Files in Unix System are organized into multi-level hierarchy structure known as a directory tree. At the very top of the file system is a directory called “root” which is represented by a “/”. All other files are “descendants” of root.
- 10) There are two main categories of I/O units in UNIX, block devices and character devices. In addition there are sockets that are used for network communication.
- 11) UNIX has a sophisticated security system that controls the ways users access files, modify system databases, and use system resources. Unfortunately, those mechanisms don't help much when the systems are misconfigured, are used carelessly, or contain buggy software.

Linux

Linux is an open source, powerful UNIX based OS. It is a 32-bit OS that runs on a variety of platforms, including Intel, SPARC, PowerPC, and DEC Alpha processors as well as multiprocessing systems. It was developed by Linux Torvalds at the University of Helsinki in Finland as a college project. He released the version of Linux, including all the source code. With the help of literally thousands of volunteers all over the world, Linux grew dramatically. It is a user-developed product, meaning that users around the world who ran the OS for their own use have developed many of its components and drivers. It is distributed through different distributors such as Red Hat, Mandrake, OpenSUSE, Ubuntu, Slakeware, Sabayon, Debian, Mandriva, PC-BSD, OpenSolaris, Gentoo, Fedora, Granular Linux, etc.

Features of Linux:

- 1) It is multi-user, multitasking, multiprogramming OS, mainly popular for server systems.
- 2) It is UNIX based OS. So, most of the features are borrowed from UNIX.
- 3) It is an open source OS.
- 4) Monolithic kernel is used.
- 5) There are multiple methods of inter-process communication under Linux: half-duplex pipes, named pipes, message queues, signals, semaphores, shared memory, memory-mapped files, sockets, and so on.
- 6) Linux uses a Completely Fair Scheduling (CFS) algorithm, which is an implementation of weighted fair queueing (WFQ). The Round Robin algorithm is generally used in time sharing environments.
- 7) The subsystem of Linux memory management is responsible to manage the memory inside the system. It contains the implementation of demand paging and virtual memory. Also, it contains memory allocation for user space programs and kernel internal structures.
- 8) Linux supports five types of files: simple/ordinary file (text file, c++ file, etc), directory, symbolic (soft) link, special file (device), and named pipe (FIFO).
- 9) Linux uses request structures to pass the I/O requests to the devices. All the block devices maintain a list of request structures.
- 10) For the basic security features, Linux has password authentication, file system discretionary access control, and security auditing. By expanding the basic standard security features it has: User and group separation, File system security, Audit trails, PAM authentication.